

to fabricate assembly. Most of the time companies do not use the right tools to save cost and end up in trouble later.

Power supply. This is one area where if SMEs compromise, the product will suffer in the field. There are two reasons for this to happen. First, they try to build the power supply in-house. This needs experience and skill, which can only be gained over a period of time and with practice. Second, they buy it from a vendor who is cheaper. A lot of times, low-cost power supplies are copy-cat designs (especially the low-cost imported ones). In any power supply, the magnetic design is the key, and if that is not mated properly with the power components, the power supply fails.

Electronics packaging or enclosure design and supply. Electronics packaging or electronics product enclosure design are the least understood and focussed areas in product design. These are always treated as boxes made out of steel or aluminium with holes for keys and connectors, and no holistic effort is ever made to address these as separate products. Electronics packaging design impacts all phases of a PLC. In addition to this, the biggest drawback in India is the lack of quality manufacturing—for electronics packaging with special materials like stainless steel and special aluminium alloys. And for manufacturing processes like stainless steel welding, aluminium welding, brazing and simple pollution-free powder coating process, established vendors are a rarity. This is one area SMEs suffer for no fault of their own and look to China, where they have perfected the art of manufacturing on a large scale at low cost. Coupled with this, the ability to design thermal solution for the products has become another issue.

Product shipment packaging. A product's success or failure is dependent on the way it is packed and sent for shipment. Any mishandling during transportation can damage the product—there have been cases of dead on arrival (DOA) due to poor packaging. Product packaging needs a holistic approach and depends on volumes of products being shipped, the size of the batch, etc. This aspect needs to be taken

care of (outsourced) from the beginning as there is little value addition in doing it in-house.

Logistics partner selection. Many times the end customer receives a product in a completely damaged state because it was not handled properly during transportation. In fact, product shipment, packaging and transportation go hand-in-hand. The most common mistake SMEs make is to confuse bulk shipment packaging with individual unit shipment packaging. This is true especially when the product is exported. When a shipment is handled in bulk through containers, products need to be packaged well. Sometimes, when individual units are shipped to end users, most SMEs use the same carton that is used for bulk shipment, which leads to damages. A multinational OEM has a very established process for testing the product in its shipment packaging, and only when it is cleared, the packaging is deemed ready for transportation. The product is dropped on all sides from a height of about 1.5 metres and is cleared only when it is not damaged.

Technical support. With an increased focus on globalisation, end customers demand that they be provided support over phone and internet during installation and usage of a product. They expect:

1. An interactive web-portal where he or she can get all the data about a product or service
2. Ability to communicate with the OEM through phone or email

Many SMEs think that displaying the product specification sheet on the web portal is sufficient to inform the customer. This forces the customer to reach out to the OEM either through email or phone. This may lead to perception problems, and the product's and OEM's credibility and brand value may take a dent.

Test and repair facility. The final challenge is to decide how to support a product that has failed in the field after prolonged use or during the warranty period. The easiest thing to do is to replace the failed product with a new one, but it is believed that if a product

fails during warranty and is replaced with a new one, it is equal to a loss of profit equivalent to five new units. In the eventuality of failure, repairing the failed product and sending it back to customer is the priority. We need to understand that when a customer pays for a product and if it fails during warranty, he expects the failed product to be replaced. If it is replaced with an old one, most probably the customer would not return. Sometimes, OEMs simply replace the external enclosure of a product to make it look like new, giving the customer a false sense of satisfaction. These OEMs think that this customer is happy and will stay. However, a repair process is time-bound and customers expect a level of service. Almost all SMEs never factor this in their planning. Today, options are available where third-party vendors undertake these repair jobs. But, proper planning is needed if this has to be outsourced, where an OEM can transfer the service level agreements to the test-and-repair partner and ensure the service quality.

Conclusion

It is a well-known fact that SMEs, though nimble, will face challenges in areas that are becoming critical and may not have in-house expertise or cannot afford expensive resources. The only choice left for them is to outsource to a capable partner. However, outsourcing needs working together and building relationships and confidence. Evaluating the outsourcing partner for cost alone will not produce desired results. The key to success in outsourcing are:

1. Having a long-term vision for the product—understanding the product's life from the eyes of the user
2. Identifying areas that need support
3. Identifying capable partners, who have skills and resources
4. Clearly documenting deliverables from both sides
5. Constantly reviewing the progress
6. Investing in long-term relationships and helping each other grow. ●

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